

GAME DESIGN DOCUMENT

Cyber Zombie Siege

Survive the waves. Upgrade your strategy. Defeat the cyber monster.



Prepared by: Islam Al-Absi and Ahmed Abu Arrah

Course Project Game Report

2026

Game Name	Cyber Zombie Siege
Game Catchphrase	Survive the waves. Defeat the cyber monster.
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Credit Page

Document Purpose: Game Design Document for Cyber Zombie Siege — a 3D first-person zombie survival shooter developed in Unity as a course project.

Document Version: v3.0 — Final Gameplay Direction

Working Title: Cyber Zombie Siege

Game Concept: A round-based first-person zombie survival shooter. The player fights two waves of increasingly difficult cyber-zombies, then faces a final Cyber Monster boss to win.

Game Document Authors: Islam Al-Absi & Ahmed Abu Arrah

Course: Game Design & Development — 2026

Sign-Off

The sign-off section confirms that all team members have read and agreed to the current game design plan.

GAME CONCEPT SIGN-OFF

Lead Designer: Islam Al-Absi Signature: IslamAlabsi Date: 9/5/2026

Lead Programmer: Ahmed Abu Arrah Signature: AhmedAbuArrah Date: 9/5/2026

Lead Artist: Islam Al-Absi Signature: IslamAlabsi Date: 9/5/2026

Lead Producer: Ahmed Abu Arrah Signature: AhmedAbuArrah Date: 9/5/2026

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Introduction

Cyber Zombie Siege is a 3D first-person survival shooter built in Unity. The player is the last survivor inside a cyber-zombie infected zone. To escape, the player must survive two waves of increasingly dangerous zombies and defeat the Cyber Monster boss controlling the outbreak. The game is original in concept, designed to demonstrate practical Unity skills: first-person controls, NavMesh enemy AI, health systems, shooting mechanics, round-based progression, runtime UI, and gameplay balancing.

Key Introduction Points	
Genre	Survival Shooter / Action
Player Type	Single Player
Technical Form	3D First-Person Perspective
Theme	Sci-Fi / Horror
Design Intention	Original — student project game
Reference	Call of Duty Zombies mode, Resident Evil

Game Analysis

Game Description

Category	Details
Genre	Survival Shooter, Action
Game Elements	Shooting, Combat, Dodging, Survival, Enemy Chase
Game Content	Sci-Fi, Horror, Thriller
Theme	Cyber-Zombie Apocalypse (Sci-Fi)
Style	Realistic 3D with dark cinematic lighting
Game Sequence	Linear — Round 1 → Round 2 → Boss Round → Victory/Game Over
Player	1 Player (Single Player only)

Game Reference

Game Taxonomy: Fictional Game/Narrative — a combat-focused game with a clear win/lose structure and a background story about a cyber-zombie outbreak.

Player Immersion: Physical (fast-paced combat), Tactical (managing health and positioning), Emotional (survival tension and boss encounter)

Reference Games: Call of Duty Zombies (round-based zombie survival), Resident Evil (horror atmosphere), Left 4 Dead (wave shooting)

Game Technical

Category	Details
Technical Form	3D Graphics
View	First-Person Perspective (FPS camera attached to player)
Platform	PC — Windows & Mac
Language	C# (via Microsoft Visual Studio)
Device	PC (Keyboard + Mouse)

Engine	Unity 3D
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Game Sales

Category	Details
Consumer Group	Gamers aged 16–30 who enjoy FPS and survival horror games
Payment	One-time purchase — digital distribution (PC storefront)
Estimated Price	\$15 USD (based on comparable indie survival shooters)

Design Document

This section describes how game objects behave, what properties they have, and how they are controlled. It defines the mechanics of Cyber Zombie Siege and is the primary reference for the game interactive systems.

Design Version

Version code: U3.0

- (U): Unity — developed in Unity 3D using C#
- (3.): Concept version 3 — final gameplay direction
- (.0): No content updates pending — final submission version

Design Guidelines

The design of Cyber Zombie Siege must follow these creative restrictions: the game must be completable in under 10 minutes; all mechanics must be achievable with a keyboard and mouse; enemy AI must use Unity NavMesh only; all UI must be built at runtime by the GameManager; difficulty must increase clearly between rounds without requiring player-read instructions mid-game.

Game Design Definitions

The following definitions establish how the core gameplay elements behave:

Menu: The main menu loads on startup with Play, Settings, and Quit options. Pressing Play leads to the mission board, then begins Round 1.

Synopsis: The player is a lone survivor in a cyber-zombie infected zone. Survive two zombie waves and defeat the Cyber Monster boss to escape.

Game Play: The player moves in first person, shoots enemies, manages health, picks up weapons, and defeats all enemies each round to advance.

Player Control: WASD to move, mouse to aim, left click to shoot, E to pick up weapon, G to drop, R to restart.

Winning: All enemies in Round 1, Round 2, and the boss are defeated. Victory screen is shown.

Losing: Player health reaches zero. Game-over panel is shown. Player can press R to restart the scene.

Game Matrix

The following table is the game matrix — all entities with their numeric properties:

- PLR_01 / Player — Health: 100, Speed: 5 units/s, Damage: weapon-dependent, Detection: N/A, Points: N/A
- ZMB_01 / Standard Zombie (Round 1) — Health: 100, Speed: 3.5, Damage: 10 per hit, Detection: 15 units, Points: 10
- ZMB_02 / Hardened Zombie (Round 2) — Health: 180, Speed: 4.5, Damage: 15 per hit, Detection: 18 units, Points: 20
- BSS_01 / Cyber Monster Boss — Health: 500, Speed: 3.0, Damage: 25 per hit, Detection: 20 units, Points: 200
- WPN_01 / Weapon Pickup — Damage: 35 per shot, Pickup Range: 2 units, Points: N/A

Game Flow Chart

The game flow follows this sequence: Launch to Main Menu to Mission Board to Round 1 to Round 2 to Boss Round to Victory or Game Over to optional Restart. [Insert flow chart diagram here]

- Menu: Start Screen with Play / Settings / Quit
- Synopsis: Mission Board shown before Round 1 begins
- Game Play: Round 1 (4 zombies) to Round 2 (7 zombies) to Boss Round
- Player Control: WASD, Mouse, Left Click, E, G, R
- Game Over (Win): Boss defeated — Victory Panel shown
- Game Over (Lose): HP = 0 — Game Over Panel — Press R to Restart

1. Design History

The design history describes the main stages of development and the changes applied to the game over time. The project began as a Unity game prototype and gradually became a first-person zombie survival shooter with a clear round-based structure, runtime user interface, enemy spawning, weapon interaction, and victory/game-over flow.

Version 1.00 - Core Prototype

1. Created the base Unity scene and established the first-person player controller.
2. Added movement, camera control, basic shooting behavior, and initial player health.
3. Imported environment assets and tested basic lighting, materials, and scene navigation.

Version 1.50 - Enemy and Combat Prototype

1. Added zombie prefabs with health, chase behavior, and attack damage.
2. Connected the zombie death event to the GameManager so the game can count remaining enemies.
3. Tested bullet damage, colliders, hit detection, and death animations.

Version 2.00 - Round System and UI

1. Implemented a GameManager that controls Round 1, Round 2, and the final boss round.
2. Added start screen, settings panel, mission board, HUD, round announcements, game-over panel, and victory panel.
3. Created runtime enemy parents to keep spawned zombies organized in the hierarchy.

Version 2.50 - Balancing and Polishing

1. Improved zombie speed, acceleration, detection range, attack damage, and round scaling.
2. Added round-based zombie health so later rounds require more shots and feel more challenging.
3. Adjusted scene lighting and fog safety settings to keep the game readable during play.

Version 3.00 - Final Gameplay Direction

1. Defined the final gameplay loop: survive two zombie waves, then defeat Cyber Monsters 2.
2. Focused on making the level playable, readable, and easy to understand during presentation.
3. Prepared documentation, screenshots, and explanation of the design decisions.

2. Game Overview

Philosophy

The goal of Cyber Zombie Siege is to create a focused and understandable survival-shooter experience. The player should immediately know what to do: move through the level, survive the zombie waves, manage health, use weapons, and defeat the final cyber monster. The design favors clarity over complexity, because the game is intended to be playable and presentable within a limited project timeline.

Common Questions

Question	Answer
What is the game?	It is a 3D first-person zombie survival shooter made in Unity. The player fights waves of zombies and then faces a final cyber monster boss.
Why create this game?	The game was created to demonstrate practical Unity skills: first-person controls, enemy AI, NavMesh movement, health systems, shooting mechanics, UI, round progression, and gameplay balancing.
Where does the game take place?	The game takes place in a dark survival arena/playground-style environment. The level is designed for fast combat, clear navigation, and zombie encounters from multiple spawn locations.
What do I control?	The player controls a first-person character who can move, aim, shoot, pick up weapons, drop weapons, and survive enemy attacks.
How many characters do I control?	The player controls one character. Enemy zombies and the boss are controlled by AI scripts.
What is the main focus?	The main focus is surviving progressively harder enemy waves and defeating the final boss.
What is different?	The game combines a simple round-based survival structure with runtime UI, automated enemy spawning, round scaling, weapon pickup behavior, and a final boss objective.

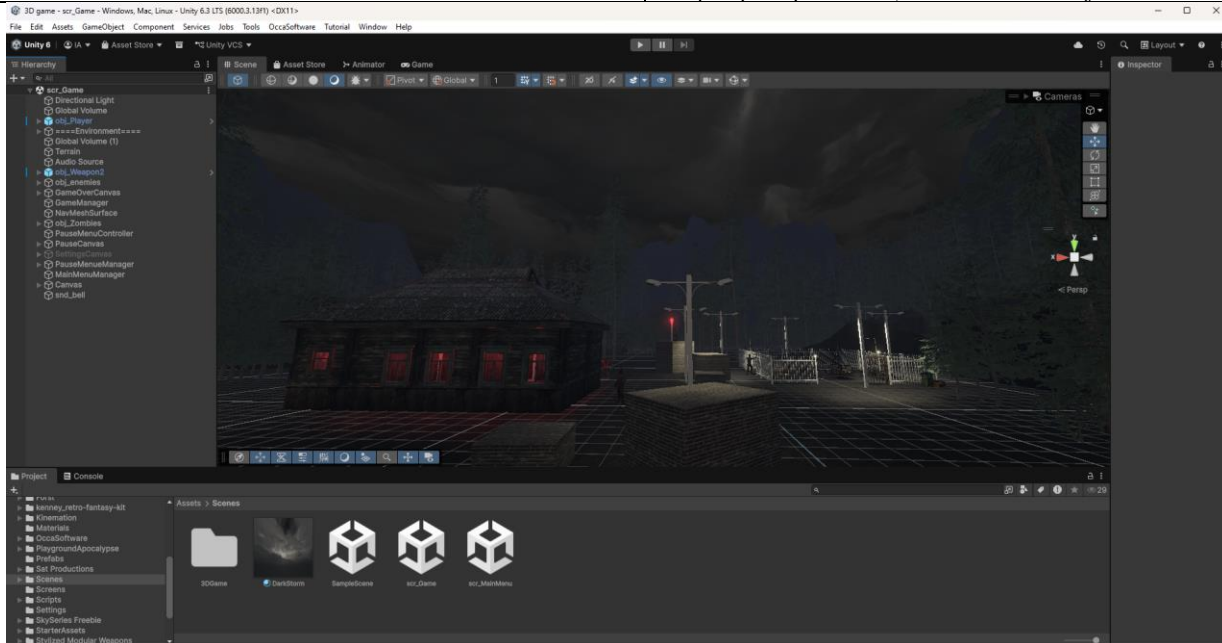


Figure 1 Unity Hierarchy

3. Feature Set

General Features

- 3D first-person gameplay built in Unity.
- Round-based enemy progression with two zombie waves and a boss round.
- Runtime menus: start screen, instructions, settings, HUD, victory, and game-over flow.
- Zombie AI with detection range, chase speed, attack damage, NavMesh movement, and death notification.
- Round scaling for zombie speed, attack damage, and health.
- Weapon pickup and weapon switching support using simple controls.
- Lighting safety system that keeps the scene visible and playable.

Single-Player Features

- One-player survival objective.
- Clear victory condition after defeating the final boss.
- Game-over state with restart support.
- Player health display on the HUD.
- Mission board explains controls and round objectives before gameplay begins.

Gameplay Features

- Round 1 introduces the player to zombie combat with fewer enemies.
- Round 2 increases pressure with more zombies and stronger stats.
- Round 3 introduces Cyber Monsters 2 as the final boss encounter.
- Zombies spawn from predefined spawn points or fallback positions if no spawn points are assigned.
- Enemies are organized under runtime parent objects in the Unity hierarchy.



Figure 2 Game Over Screen

Game Atmosphere

This section describes the overall look, feel, and audio direction of Cyber Zombie Siege. The atmosphere is dark sci-fi horror: threatening and tense, while visually readable enough for the player to aim and react.

Atmosphere Mood Board

Visual references: dark industrial environments, neon-tinted lighting, foggy corridors. Color palette is desaturated dark grey, rust, and teal-cyan accents. Enemy design features cyber-organic creatures with glowing eyes. HUD uses clean white-on-dark minimal UI.

Character / Units Sketch and Description

Player: invisible in first person; only the held weapon is visible. Zombies: humanoid figures in torn clothing with cyber implants, glowing eyes, and decomposed skin. Cyber Monster Boss: larger than standard zombies, heavily armored, slower but more powerful. [Insert character sketches here]

Level (Locations) Sketch and Description

The arena is a contained outdoor/indoor hybrid space. The central open area allows combat and repositioning. Perimeter structures provide cover and zombie spawn zones. Fog and darkness limit long-range visibility, increasing tension. [Insert level layout sketch or overhead screenshot here]

Audio Description

Ambient: low industrial hum, distant zombie growls, wind. Combat: weapon fire, zombie snarls, footsteps. Boss encounter: escalating music cue, heavy footsteps. Victory: short resolution sting. Game Over: low descending tone.

Selling Features

The following features are the primary marketing and selling points of Cyber Zombie Siege:

- 3 rounds of increasing difficulty with a distinct boss encounter
- Fast-paced first-person zombie survival — approx. 5 to 10 minutes per run
- High replayability due to random spawn points and round-scaled enemy stats
- Built in Unity 3D with NavMesh AI, runtime UI, and scalable enemy stats
- Single player with clear win/lose conditions and restart support
- Target consumer: PC gamers aged 16 to 30 who enjoy survival shooters
- Unique feature: lighting safety system ensures playability in dark environments
- Scoring system with per-kill points, round bonuses, and live HUD display

4. The Game World

Overview

The game world is a compact 3D combat space. It is not designed as a large open world; instead, it is designed as a functional arena where the player can quickly understand enemy direction, movement routes, and survival

pressure. The map supports multiple zombie spawn locations so the player feels surrounded without becoming lost.

World Feature #1 - Combat Arena

The arena gives the player enough space to move, aim, and reposition. The layout should include open areas for shooting, cover or obstacles for tension, and clear paths for NavMeshAgents so zombies can reach the player reliably.

World Feature #2 - Horror/Cyber Atmosphere

The visual direction uses a dark survival mood with cyber-zombie enemies. Lighting, fog, and environment props should create tension while preserving visibility. The game should look dangerous but still be readable during gameplay.

5. The Physical World

Overview

The physical world is built around movement, collision, shooting, and enemy navigation. Unity colliders define what the player and enemies can touch, while NavMesh allows enemies to chase the player across walkable areas.

Key Locations

- Player start area: the location where the player begins the match.
- Zombie spawn zones: points around the map where enemies appear each round.
- Boss spawn zone: the location where the Cyber Monster boss enters the game.
- Weapon pickup positions: places where the player can obtain or switch weapons.
- Combat routes: walkable paths that allow zombies to chase the player.

Travel and Scale

The player travels on foot using first-person movement. The scale is practical for a student project: large enough for combat and repositioning, but small enough to keep objectives clear. Zombies use NavMeshAgent movement, so the walkable areas must be baked correctly.

Objects, Weather, Day/Night, and Time

Objects include environmental props, weapons, zombies, UI objects, lights, and spawn point markers. The game does not require dynamic weather or a real time-of-day cycle. A fixed dark/cinematic lighting setup is more suitable because it keeps the mood consistent and reduces implementation risk.

6. Rendering System

The game uses Unity 3D rendering with a first-person perspective. The focus is stable performance, readable enemy silhouettes, and consistent material rendering. Imported assets may require material fixes, especially if they appear pink because of shader pipeline mismatches.

2D/3D Rendering

The world, enemies, weapons, and environment are rendered in 3D. The user interface is rendered as screen-space overlay canvas objects. This allows the game to display HUD elements such as health and enemy count without being affected by camera movement.

Performance Considerations

- Use optimized low-poly or medium-poly assets for enemies and props.
- Avoid placing too many dynamic lights in the scene.
- Bake the NavMesh carefully so zombies do not get stuck.
- Reduce unnecessary imported assets from the project to keep build size manageable.
- Test game view and scene view separately because editor lighting can look different from runtime lighting.

7. Camera

The camera is a first-person camera attached to or controlled by the player. The camera must align with the player's aiming direction so shooting feels responsive and natural.

Camera Detail #1 - First-Person Control

The player uses mouse movement to rotate the camera and WASD to move. The camera stays near the player's head position and gives the player direct control over aim and awareness.

Camera Detail #2 - Gameplay Readability

The camera should avoid excessive effects that reduce visibility. Cinematic lighting is useful, but the player must still clearly see enemies, bullets, pickups, and the environment.



Figure 3 Game view screenshot

8. Game Engine

Overview

The game is built in Unity, a cross-platform engine used for 2D and 3D games. Unity provides the main systems needed by this project: scenes, prefabs, components, physics, colliders, UI canvases, lighting, audio, input, NavMesh, and C# scripting.

Game Engine Detail #1 - Scripting

C# scripts define the game flow. The GameManager controls the round sequence, spawning, UI panels, victory, and game over. ZombieHealth manages health, damage, death, and notification to the GameManager. ZombieAI controls enemy chasing and attacking.

Collision Detection

Unity colliders are used for bullets, enemies, player body, pickups, and obstacles. Zombie colliders are disabled after death so bullets and physics stop interacting with dead enemies. This avoids repeated damage and incorrect round counting.

Navigation

Zombies use NavMeshAgent to follow the player. Spawn points should be placed on or near baked NavMesh areas. If a spawn position is not on the NavMesh, the GameManager samples a nearby NavMesh position to improve reliability.

9. Lighting Models

The game uses a readable dark atmosphere. Lighting should make enemies visible while keeping the survival mood. The GameManager includes a lighting safety feature that increases ambient light, controls fog density, and ensures that at least one directional light exists.

Lighting Model Detail #1 - Directional and Ambient Light

A directional light acts like moonlight or distant outdoor lighting. Ambient lighting prevents the scene from becoming completely black. This is important because the player needs to aim at moving zombies.

Lighting Model Detail #2 - Point and Spot Lights

Point and spot lights can be used for street lights, torches, weapon effects, or environmental highlights. They should be used carefully because too many real-time lights can hurt performance. If a light appears not to affect the scene, its range, intensity, render pipeline compatibility, and shadow settings should be checked.



Figure 4 Game Lighting

10. The World Layout

Overview

The world layout should support a clear sequence of combat. The player starts in a central or semi-protected position, zombies spawn from multiple directions, and the boss appears after the second wave. The layout must balance challenge with fairness.

World Layout Detail #1 - Map Plan

The map is designed as an open combat arena that supports wave-based zombie survival gameplay. The player starts near the center of the map in a semi-protected area, giving enough time to understand the mission, move, aim, and prepare for the first wave. The environment includes open walking paths, cover objects, weapon pickup locations, and several enemy spawn zones around the map.

Zombies do not spawn from one fixed position. Instead, they spawn randomly from multiple predefined spawn points around the arena. This makes each round feel less predictable and forces the player to stay alert, check different directions, and keep moving during combat. The spawn points are placed far enough from the player to avoid unfair instant attacks, but close enough to create pressure.

World Layout Detail #2 - Challenge Flow

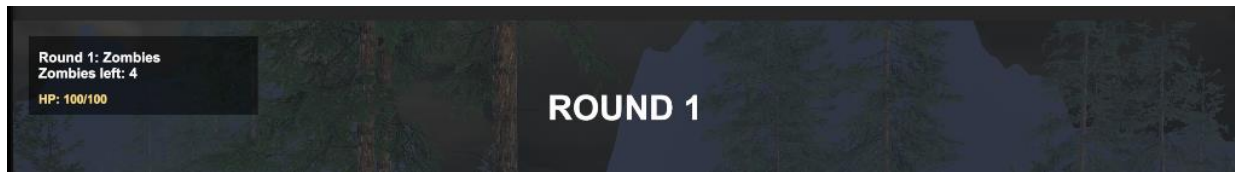


Figure 5 Photo of showing Round 1, Round 2, and status bar (health and zombies left)

The map should show clear level design planning, including enemy spawn positions, player routes, weapon locations, and the location of the final boss. This directly supports the documentation expectations in the project rubric.

11. The Gameplay

Overview

The gameplay is based on surviving enemy waves. The player starts the game from the menu, reads the mission board, enters Round 1, clears the first zombie wave, clears a harder second wave, and then defeats the Cyber Monsters 2 boss. The game ends with victory or game over depending on whether the player survives.

Actions of the Gameplay

- Move using WASD.
- Aim using the mouse.
- Shoot using left click.
- Pick up weapons using E.
- Drop weapons using G.
- Survive zombie attacks and avoid losing all health.
- Press R to restart after game over.

Hierarchy of Challenges

Win the Game
Defeat Round 3 Boss
Clear Round 2 Zombie Wave
Clear Round 1 Zombie Wave
Survive and Manage Health
Move, Aim, Shoot, Pick Up Weapon

Table: Hierarchy of challenges for the single-player zombie survival mode.

12. The Core Mechanics

Overview

The core mechanics are intentionally clear and easy to understand. The player shoots enemies, enemies chase and attack the player, and the GameManager tracks whether all enemies in the current round have been defeated.

Rules of the Game Detail #1 - Round Progression

1. Round 1 starts with a smaller number of zombies.
2. When all Round 1 zombies are killed, the GameManager waits a few seconds and starts Round 2.
3. Round 2 spawns more zombies with stronger health, speed, and attack damage.
4. When all Round 2 zombies are killed, the boss round begins.
5. When the boss is killed, the victory screen appears.

Rules of the Game Detail #2 - Enemy Scaling

Round	Enemy Count	Zombie Health	Gameplay Effect
Round 1	4 zombies	100 HP	Introduction wave; teaches combat basics.
Round 2	7 zombies	180 HP	Harder survival wave; more enemies and more pressure.
Round 3	Boss	500 HP	Final encounter; victory depends on defeating the boss.

Rules of the Game Detail #3 - Game Over

If the player loses all health, the GameManager shows the game-over panel, unlocks the cursor, disables player controls, and optionally pauses the game. The player can restart the scene after game over.

Rules of the Game Detail #4 - Scoring and Level Progression

Cyber Zombie Siege uses a round-based scoring system. Players earn points by defeating enemies and completing rounds. The scoring system is designed to be obvious to the player at all times through the HUD display.

Score Values:

- Each standard zombie killed: +10 points
- Completing Round 1 without losing a life: +50 bonus points
- Each Round 2 zombie killed: +20 points (stronger enemies worth more)
- Defeating Cyber Monster Boss: +200 points
- Surviving with full health at end of any round: +25 bonus points

The current score is displayed on the HUD at all times. Round completion is the primary progression gate: the player cannot advance to Round 2 until all Round 1 zombies are defeated, and the boss round does not begin until Round 2 is cleared. This makes progression obvious and directly tied to player performance.

Rules of the Game Detail #5 - Level Design and Difficulty Progression

The game contains three distinct levels (rounds), each increasing in difficulty. All three levels are winnable with correct strategy, but each one requires more skill and awareness than the previous.

Level 1 (Round 1) - Introduction: 4 zombies with 100 HP each. Low speed. The player learns movement, shooting, and weapon pickup. Forgiving pacing to establish core mechanics. Winnable easily once controls are understood.

Level 2 (Round 2) - Escalation: 7 zombies with 180 HP each. Higher speed and attack damage. The player must manage health, positioning, and weapon usage more carefully. Spawn points increase pressure from multiple directions.

Level 3 (Boss Round) - Final Encounter: Cyber Monsters 2 boss with significantly higher health and different attack behavior. The player must apply all learned skills. Defeating the boss triggers the victory screen and is the win condition for the game.

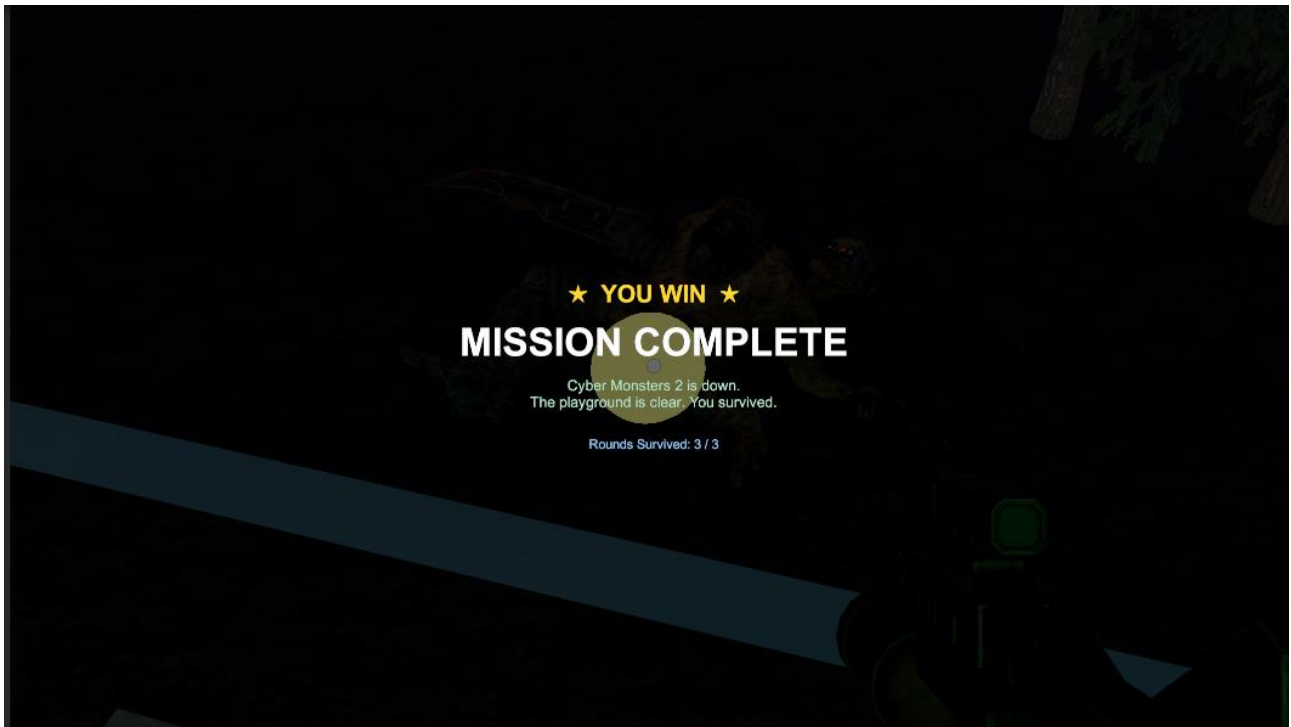


Figure 6 Winning screen

13. Data Structure of Entities and Relationships

The following table summarizes the main entities in the game and how they interact. This helps explain the implementation design behind the gameplay.

Entity	Main Attributes	Behavior	Round Scaling	Relationship
Player	Health, movement, camera, weapon holder	Moves in first person, shoots, picks up/drops weapons	No round scaling	Attacked by zombies and boss
Zombie	Health, damage, chase speed, detection range	Uses NavMeshAgent and ZombieAI to chase and attack player	Health/speed/damage increase by round	Spawned by GameManager
Cyber Monster Boss	Boss health, detection range, attack behavior	Final enemy after round 2 is cleared	Appears in boss round	Victory condition depends on death
Weapon Pickup	Weapon name, pickup distance, ammo/weapon type	Can be equipped with E and dropped with G	Static	Used by player to damage enemies
GameManager	Rounds, enemy counts, UI, spawning, victory/game over	Controls game flow and spawned enemies	Applies scaling during spawn	Receives death notifications

Game Flow Relationship

The GameManager is the central controller. It spawns enemies, prepares their health and AI values, counts enemies remaining, advances rounds, and shows victory or game-over UI. ZombieHealth communicates death back to the GameManager. ZombieAI controls chasing and attacking during gameplay.

14. Game Characters

Player Character

The player character is the survivor controlled from a first-person perspective. The player is the only playable character and must survive the zombie siege using movement, aiming, shooting, and weapon pickups.

Cyber Monster Boss

The Cyber Monster is the final enemy and represents the highest challenge in the current version. It appears after the two zombie rounds are cleared. The boss fight gives the game a final objective instead of ending after ordinary waves.

Character Rendering and Animation

Zombie characters use animations such as idle, walk, run, attack, falling back, and death. Animation transitions must be configured carefully so attack and death animations can trigger from the correct states. Root motion should usually be disabled when NavMeshAgent controls movement.



Figure 7 Cyber Monster Boss

15. Enemies and Monsters

Zombies

Zombies are the main enemy type. They are spawned by the GameManager during Round 1 and Round 2. Each spawned zombie receives the current round values for health, speed, and attack damage.

AI Behavior

- Detect the player within a minimum detection range.
- Chase the player using NavMeshAgent movement.
- Stop near the player using stopping distance.
- Trigger attack behavior when close enough.
- Notify the GameManager when killed.

Known Enemy Setup Notes

- Spawn points should be placed on the NavMesh or close enough for NavMesh.SamplePosition to correct the location.
- Round 2 Zombie 5 and Zombie 6 may reuse earlier spawn points if fewer than seven spawn points are assigned.
- Animator parameters must match script names exactly, such as Speed, Attack, and Die.
- Any State to Attack transition is recommended so zombies can attack from idle, walk, or run.

Player Elements

This section defines all elements directly related to the player in Cyber Zombie Siege. The player is referred to internally as PLR_01 and in-game as The Survivor.

Player Definition

- Default Status: Player starts at 100 HP, positioned at the center spawn point, unarmed until a weapon pickup is collected.
- Actions: Move, aim, shoot, pick up weapon, drop weapon, restart after game over.
- Information (HUD): Current health, current round number, number of zombies remaining.
- Default Properties: 100 HP, no weapon equipped, first-person camera active, cursor locked.
- Winning: Defeat all enemies in Rounds 1, 2, and the boss in Round 3.
- Losing: Health reaches zero — game-over panel shown.

Player Properties

Each property below defines the attribute, its range, and the feedback shown when it changes:

- Health — Range: 0 to 100. Feedback: HUD health value updates in real time. Red screen flash on damage. Game-over panel triggered when health reaches 0.
- Weapon Equipped — State: None or weapon name. Feedback: Weapon model appears in player hands. Shooting becomes available when weapon is held.
- Movement Speed — Value: 5 units per second. Feedback: WASD input moves player in real time with no delay.
- Mouse Sensitivity — Configurable in Settings panel. Feedback: Camera rotation speed changes immediately when adjusted via slider.

Player Rewards — Power-ups and Pick-ups

The following objects benefit the player when collected:

- Weapon Pickup (WPN_01): Equips the player with a shooting weapon. Activated by pressing E when within 2 units. Effect: player can now shoot and deal damage. Feedback: weapon model appears in hands, fire sound plays.
- Score Bonus — Round Completion: +50 points for completing Round 1 without dying; +25 points for surviving any round with full health. Displayed on HUD immediately.

Note: there are no health pickups in the current version. Players must preserve their starting 100 HP across all three rounds.

Antagonistic Definitions

An antagonistic element in Cyber Zombie Siege is any enemy-controlled entity that detects the player, chases the player, and deals damage on contact. All antagonistic elements are spawned and managed by the GameManager. They cannot be controlled by the player.

Antagonistic Properties

All antagonistic elements share these properties. Values differ by type and round:

- Health (HP): Amount of damage the enemy can absorb before dying. Increases by round.
- Chase Speed: NavMeshAgent movement speed toward the player.
- Attack Damage: HP removed from player per successful hit.
- Detection Range: Radius in Unity units within which the enemy detects the player.
- Stopping Distance: Distance at which the enemy stops moving and begins attacking.
- Death Notification: On death, ZombieHealth calls GameManager.EnemyKilled() to update round count.

Antagonistic List

- ZMB_01 — Standard Zombie: round 1 enemy. 100 HP, speed 3.5, damage 10 per hit, detection 15 units. Humanoid model with idle, walk, attack, and death animations.
- ZMB_02 — Hardened Zombie: round 2 enemy. 180 HP, speed 4.5, damage 15 per hit, detection 18 units. Same model class as ZMB_01 with scaled stats.
- BSS_01 — Cyber Monster Boss: round 3 only. 500 HP, speed 3.0, damage 25 per hit, detection 20 units. Larger model, distinct attack animation.

Artificial Intelligence (AI)

Each antagonistic element uses the following four-state AI model, implemented via ZombieAI.cs and NavMeshAgent:

- Normal State: Enemy idles at spawn position when no player is detected. NavMeshAgent speed is set to patrol speed (1.0). Idle or walk animation plays.
- Detection State: Player enters detection range (15 to 20 units depending on type). ZombieAI detects player via distance check each frame. Chase behavior activates.
- Reaction State: Enemy chases player at full chase speed using NavMeshAgent.SetDestination. Walk or run animation plays. Stops at stopping distance (1.5 units).
- End State: Player defeats enemy — ZombieHealth reduces HP to 0, triggers Die animation, disables collider, and calls GameManager.EnemyKilled(). Enemy object deactivates after death animation completes.

16. User Interface

Overview

The user interface is built at runtime by the GameManager. It uses a screen-space canvas and contains all major game flow panels. This approach allows the game to run even if UI objects are not manually placed in the scene.

User Interface Detail #1 - Menus

- Start screen with Play, Settings, and Quit buttons.
- Settings panel with volume and mouse sensitivity sliders.
- Instructions or Mission Board panel explaining the round structure and controls.
- Victory panel after the boss is defeated.
- Game-over panel when the player dies.

User Interface Detail #2 - HUD

The HUD displays the current round, the number of zombies remaining, and player health. This keeps the player informed about progress and survival status during combat.



Figure 8 Pause Screen UI



Figure 9 zombie

17. Weapons

Overview

Weapons are used by the player to damage zombies and the boss. The game supports weapon pickup behavior so the player can interact with weapons placed in the scene.

Weapon Detail #1 - Shooting Weapon

The shooting weapon fires at enemies using the player aim direction. Damage is applied to ZombieHealth or boss health scripts. The weapon should have clear feedback such as sound, muzzle flash, bullet visuals, or hit effects.

Weapon Detail #2 - Weapon Pickup

Weapon pickups allow the player to equip weapons using the E key when close enough. The game also supports dropping weapons with G. Weapon names are listed on the mission board if pickups exist in the scene.



Insert screenshots of weapon pickup, equipped weapon, bullet effect, and hit effect

18. Musical Scores and Sound Effects

Overview

Sound effects improve feedback and immersion. The game benefits from weapon fire sounds, zombie attack sounds, zombie death sounds, footstep sounds, menu clicks, and victory/game-over sounds.

3D Sound

Enemy sounds can use 3D spatial audio so the player can hear where zombies are coming from. This is especially useful in a first-person survival game because enemies may approach from behind or from the side.

Sound Design

- Gunshot sound for shooting feedback.
- Zombie growl and attack sound for threat feedback.
- Footstep or running sound for player movement.
- Pickup sound for weapons.
- Round start and boss incoming sound cues.
- Game-over and victory sound cues.

Global Game Elements

This section describes the world boundaries, neutral objects, camera behavior, and scale used throughout Cyber Zombie Siege.

- World Boundaries: Invisible collider walls surround the combat arena. Players and enemies cannot leave the playable area. NavMesh bake defines the walkable region.
- Neutral Objects: Environmental props (barrels, crates, fences, light posts) that do not interact with player or enemy scripts. They provide cover and visual detail only.
- Camera View: First-person camera attached to the player head position. Camera rotates with mouse input.
- World Scale: The arena is sized for approximately 30 seconds of cross-map travel at player speed. This keeps combat focused and ensures zombies reach the player within a reasonable time.
- Lighting: Directional light simulates moonlight. Ambient light prevents complete darkness. GameManager includes a lighting safety check on startup.
- Fog: Enabled at medium density to limit long-range visibility and increase tension.

The Story Copy — In-Game Version

The following is the short in-game story text displayed on the Mission Board screen before Round 1 begins:

A cyber-zombie outbreak has overrun the zone. You are the last survivor. Two waves of infected stand between you and escape. Kill them all. Survive the boss. Get out.

Round objectives shown on mission board:

- Round 1: Eliminate 4 zombies to advance.

- Round 2: Eliminate 7 hardened zombies to advance.
- Round 3: Defeat the Cyber Monster boss to win.

Concept Art

This section contains visual references and concept sketches for Cyber Zombie Siege. Replace each placeholder below with an actual screenshot or sketch in the final submission.

- [Placeholder: Player perspective screenshot showing held weapon and HUD]
- [Placeholder: Zombie enemy model front view]
- [Placeholder: Cyber Monster Boss model]
- [Placeholder: Level overhead sketch with spawn points marked]
- [Placeholder: Main menu UI mockup]
- [Placeholder: HUD layout showing health, round counter, enemy count]

Creative restriction: all assets must be compatible with Unity URP or Built-in Render Pipeline. Imported assets with pink materials must have shaders reassigned before submission.

Level Copy

The following is the in-game text displayed during level transitions, shown via UI panels managed by the GameManager:

Round 1 Start: Round 1 — Survive the first wave. 4 zombies incoming.

Round 2 Start: Round 2 — More are coming. Stay alive. 7 zombies incoming.

Boss Round Start: FINAL ROUND — The Cyber Monster has arrived. Defeat it to escape.

Victory: You survived. The outbreak is over. Mission complete.

Game Over: You were overwhelmed. Press R to try again.

Game Architecture

The game architecture describes the full screen flow of Cyber Zombie Siege. Each screen is numbered below for reference.

- Screen 01 — Title / Splash Screen: Game name, tagline, and logo displayed on startup.
- Screen 02 — Main Menu: Buttons for Play, Settings, and Quit.
- Screen 03 — Settings Panel: Volume slider and mouse sensitivity slider.
- Screen 04 — Mission Board / Instructions: Round objectives and full control list.
- Screen 05 — Round 1 HUD: Health, round label, zombie count. Active during gameplay.
- Screen 06 — Round 2 HUD: Same layout; zombie count updates to 7.
- Screen 07 — Boss Round HUD: Boss health bar added. Round label shows Boss Round.
- Screen 08 — Victory Panel: Mission Complete text, final score, and main menu button.
- Screen 09 — Game Over Panel: Game Over text, current score, and restart prompt (Press R).

Game Architecture Overview

Splash screens and menus follow the game dark sci-fi style. The most important options (Play, Restart) are accessible in one click or one key press. Instructions are written in plain language so any player understands the rules without prior experience. The current session score is always visible on the HUD. A final score is shown on both the victory and game-over screens.

Architecture Copy

All in-game text compiled here for reference:

- Main Menu: Cyber Zombie Siege | Buttons: Play, Settings, Quit
- Settings: Volume [slider 0 to 100] | Mouse Sensitivity [slider 0 to 100]
- Mission Board: Round 1: Kill 4 zombies. Round 2: Kill 7 zombies. Boss: Defeat the Cyber Monster. Controls: WASD Move | Mouse Aim | Left Click Shoot | E Pick Up | G Drop | R Restart
- HUD: HP: [value] | Round: [number] | Enemies: [count]
- Victory: Mission Complete! Score: [value] | Button: Main Menu
- Game Over: Game Over. Score: [value]. Press R to Restart.

How To Play Copy

The following instructions are shown to the player on the Mission Board screen:

- Move using W, A, S, D.
- Aim using the Mouse.
- Shoot using Left Mouse Button.
- Pick up a weapon by walking near it and pressing E.
- Drop your current weapon by pressing G.
- Kill all zombies in each round to advance.
- Do not let your health reach zero.
- Defeat the Cyber Monster boss in Round 3 to win.
- If you die, press R to restart from Round 1.

19. Single-Player Game

Overview

The single-player mode is the main mode. The player must survive through the full sequence of rounds. It is designed to be completed in a short gameplay session suitable for demonstration and grading.

Story

A cyber-zombie outbreak has taken over the area. The player is the last survivor inside the infected zone. To escape, the player must clear two waves of zombies and defeat the Cyber Monster controlling the outbreak.

Hours of Gameplay

The expected gameplay length for one successful run is approximately 5 to 10 minutes depending on player skill, weapon damage, zombie health values, and map size.

Victory Conditions

- Clear all zombies in Round 1.
- Clear all zombies in Round 2.
- Defeat the Cyber Monsters 2 boss in Round 3.
- Reach the victory screen without dying.

20. Testing and Known Issues

Testing Performed

- Round progression tested from Round 1 to Round 2 and boss round.
- Zombie spawning tested using GameManager-created runtime enemies.
- Zombie health scaling tested using SetHealth on spawned ZombieHealth components.
- Zombie movement tested with NavMeshAgent speed, acceleration, angular speed, and stopping distance.
- UI tested for start screen, mission board, HUD, game over, and victory.
- Restart behavior tested after game over.

Known Issues and Fixes

Issue	Cause	Fix / Recommendation
Round 2 Zombie 5/6 spawn behavior	Spawn points may be reused if fewer than 7 points exist	Add at least 7 spawn points or adjust spawn logic.
Zombie attack animation not playing	Animator transition may not allow attack from all states	Add Any State -> Attack transition using Attack trigger.
Zombies feel slow	Low chase speed or NavMeshAgent acceleration	Increase round speed, acceleration, and angular speed.
Imported material appears pink	Shader/render pipeline mismatch	Convert materials or assign compatible shaders.
Scene view looks darker than Game view	Scene lighting settings differ from runtime camera/game lighting	Check Scene Lighting toggle and post-processing settings.

21. Group Work and Contributions

The project was divided into practical development responsibilities to reduce conflicts and allow each member to focus on a clear system. Since Unity projects can create merge conflicts, team members should avoid editing the same scene at the same time and should communicate before changing shared prefabs or core scripts.

Member / Role	Main Contribution	Evidence to Add
Gameplay / GameManager	Rounds, spawning, UI flow, victory/game-over, scaling	Screenshots of GameManager inspector and scripts
Enemy AI / Animation	Zombie chase, attack behavior, health, death animations	Screenshots of Animator and enemy prefab setup
Level Design / Art	Map layout, lighting, props, spawn points, navigation	Screenshots of level map and Scene view
Weapons / Audio	Weapon pickup, shooting, bullet visuals, sounds	Screenshots of weapon prefabs and test gameplay

22. Appendices

Appendix A - Controls

Input	Action
WASD	Move
Mouse	Aim / Look
Left Click	Shoot
E	Pick up weapon
G	Drop weapon
R	Restart after game over

Appendix B - Screenshot Checklist

- Main menu or title screen.
- Mission board / instructions screen.
- Round 1 zombie wave gameplay.
- Round 2 zombie wave gameplay.
- Boss fight screenshot.
- Map overview with spawn points marked.
- Animator controller screenshot.
- GameManager inspector screenshot.
- Victory screen and game-over screen.

Appendix C - Final Notes

This document should be updated after final screenshots are added. Replace each marked placeholder with a real screenshot, and keep the caption below each image to explain what the screenshot demonstrates.

Appendix D - Naming Conventions

All game assets in Cyber Zombie Siege use consistent naming conventions with prefixes so any team member can identify the asset type immediately. The following prefix system is applied throughout the Unity project.

Asset Prefix System:

- **spr_** — Sprites and 2D textures. Example: spr_PlayerHUD, spr_HealthIcon, spr_CrosshairDot

- **snd_** — Sound effects and audio clips. Example: snd_GunShot, snd_ZombieGrowl, snd_ZombieDeath, snd_WeaponPickup, snd_Victory, snd_GameOver
- **back_** — Background images and skybox materials. Example: back_MainMenu, back_NightSky, back_DarkArena
- **scr_** — C# scripts and controllers. Example: scr_GameManager, scr_ZombieAI, scr_ZombieHealth, scr_PlayerMovement, scr_WeaponPickup, scr_BossHealth
- **font_** — Font assets. Example: font_HUDMain, font_MenuTitle, font_ScoreDisplay
- **obj_** — GameObjects and prefabs. Example: obj_ZombiePrefab, obj_BossPrefab, obj_PlayerController, obj_WeaponPickup, obj_BulletPrefab, obj_SpawnPoint
- **room_** — Scenes and rooms. Example: room_MainMenu, room_Round1, room_Round2, room_BossArena, room_Victory

All members of the team follow this naming system when importing assets, creating prefabs, naming scripts, and naming scenes. This ensures the Unity project folder remains navigable and any reviewer can quickly locate asset types by inspecting their name prefix.

Appendix E - Game Info Screen Requirements

The in-game Info / Mission Board screen includes all required information a player needs before starting. It is accessible from the main menu and also from the pause menu during play.

Game Name: Cyber Zombie Siege — displayed prominently on the title screen and mission board.

Author Name: Islam Al-Absi and Ahmed — shown on the title screen credits.

How to Play: WASD to move, Mouse to aim, Left Click to shoot, E to pick up weapon, G to drop weapon, R to restart. Instructions listed on the Mission Board screen.

Rules: Kill all zombies to advance each round. Do not let your health reach zero. Defeat the Cyber Monster boss in Round 3 to win.

Technical Document

This section covers the technical requirements and constraints for Cyber Zombie Siege, organized by category: system requirements, visual content, audio content, programming content, code structure, concerns, resources, and technical matrix.

System Requirements

Minimum requirements to run Cyber Zombie Siege:

- OS: Windows 10 or macOS 10.14 or later
- CPU: Intel Core i5 or equivalent (2.0 GHz or higher)
- RAM: 4 GB minimum, 8 GB recommended
- GPU: Dedicated GPU with OpenGL 4.5 or DirectX 11 support

- Storage: Approximately 500 MB for the final build
- Input: Keyboard and mouse required
- Display: 1280x720 minimum resolution

Visual Content

Technical requirements for all visual and graphical assets:

- General — File format: PNG for textures, FBX for 3D models. Max texture size: 1024x1024 for environment, 512x512 for HUD icons. Render pipeline: Built-in or URP, consistent across all assets.
- Player Elements — States: Default (first-person weapon held), Damaged (red screen flash), Dead (camera drop and freeze). Animation: weapon idle, shoot, reload animations only.
- HUD — Icons: health icon, round counter label, enemy count label. Font: bold sans-serif (Arial). Colors: white on semi-transparent dark background.
- Antagonistic Elements — States: Idle, Walk, Run, Attack, Damaged, Death. Min 12 animation frames per state. Collider disabled after death animation frame 8 or later.
- Global Elements — Background/Sky: dark night sky texture or solid dark color. Environment textures: tileable ground, wall, prop textures at 512x512. Font: Arial for all UI.

Audio Content

Technical requirements for all audio assets:

- General — File format: WAV or OGG. Sample rate: 44100 Hz. Bitrate: 128 kbps minimum for music, 96 kbps for SFX. SFX files under 500 KB, music loops under 5 MB.
- Player Elements — Sounds: footstep (looping), weapon fire (one-shot), weapon pickup (one-shot). Platform: PC only, no vibration.
- Antagonistic Elements — Sounds: zombie growl (loop during chase), zombie attack (one-shot), zombie death (one-shot), boss roar (trigger on spawn).
- Global Elements — Ambient music: low tension loop for gameplay. Loops seamlessly. 60 to 80 BPM.
- Splash Screens — Short intro sting on game load (2 to 3 seconds).
- Menus — Sound: button click (one-shot on interactions). Slider drag: no sound.

Programming Content

Technical requirements for the programming implementation:

- General — Language: C# (Unity scripting API). Unity version: 2021 LTS or later. Convention: PascalCase for classes, camelCase for variables. No external plugins required.
- Player Events — Triggers: OnTriggerEnter for weapon pickup, Update for shooting input, OnDeath when health reaches zero.
- Antagonistic Events — Triggers: distance check in Update for detection, NavMeshAgent.SetDestination per frame in chase, OnDeath notifies GameManager.EnemyKilled().
- Global Events — NavMesh bake required before build. Lighting safety check on GameManager.Start(). Fog density set programmatically if missing.
- Splash Screens — Event: scene loaded, intro sting plays, logo shown 2 seconds, then Main Menu displays.
- Menus — Events: button onClick listeners registered at runtime. Settings sliders use onValueChanged.

Code Structure

Overview of how scripts interact in the game:

- GameManager.cs: central controller. Owns round state, spawn logic, enemy count, UI panels, victory and game-over flow. All other scripts report to GameManager.
- ZombieHealth.cs: tracks enemy HP, applies bullet damage, triggers death animation, calls GameManager.EnemyKilled() on death.
- ZombieAI.cs: reads player position, manages NavMeshAgent, controls detection and attack state machine.
- PlayerMovement.cs: handles WASD movement and mouse look input.
- PlayerShooting.cs: handles raycast or bullet instantiation, applies damage to hit targets.
- WeaponPickup.cs: detects E key input within pickup range, equips weapon to player.
- Module write order: PlayerMovement, PlayerShooting, WeaponPickup, ZombieHealth, ZombieAI, GameManager, then UI panels.

Concerns and Alternatives

- Concern: NavMesh spawn placement — enemies may spawn off-mesh. Alternative: use NavMesh.SamplePosition to correct spawn point before instantiation. Already implemented in GameManager.
- Concern: Material shader mismatch on asset import (pink materials). Alternative: convert materials to Built-in Standard shader or reassign manually in Inspector.
- Concern: Round 2 spawn points fewer than 7. Alternative: GameManager wraps spawn index, reusing earlier positions when fewer points are assigned.
- Concern: Animation parameter name mismatch. Alternative: verify Animator parameter names (Speed, Attack, Die) match ZombieAI script string references exactly.

Resources

Applications and assets approved for use in the development of this game:

- Unity 3D — game engine (unity.com)
- Microsoft Visual Studio or VS Code — C# IDE
- Unity Asset Store — free enemy and environment asset packs (Unity Asset Store EULA)
- Mixamo — free zombie animations (Adobe Mixamo standard license)
- Freesound.org — royalty-free SFX (Creative Commons)
- Unity NavMesh — built-in pathfinding, no additional license required

Technical Matrix

Content lists by category for all three asset types:

- Audio: weapon_fire.wav, zombie_growl.ogg, zombie_attack.wav, zombie_death.wav, boss_roar.wav, ambient_loop.ogg, victory_sting.wav, gameover_sting.wav, button_click.wav
- Visual: spr_HealthIcon.png, spr_CrosshairDot.png, font_HUDMain.ttf, env_GroundTile.png, env_WallTile.png, obj_ZombiePrefab.fbx, obj_BossPrefab.fbx
- Programming: scr_GameManager.cs, scr_ZombieAI.cs, scr_ZombieHealth.cs, scr_PlayerMovement.cs, scr_PlayerShooting.cs, scr_WeaponPickup.cs, scr_BossHealth.cs

Scoring: +10 per zombie (Round 1), +20 per zombie (Round 2), +200 for boss kill, +50 bonus for a perfect round, +25 for surviving with full health. Score displayed live on the HUD.